TABLE I
DIFFERENT TYPES OF OLED DISPLAYS

OLED type	Working	Features	Applications
Passive-matrix OLED (PMOLED)	It consists of organic layer and strips of anode and cathode that are perpendicular to each other. This intersection makes up the pixels and light is emitted.	• PMOLED displays are easy and cheap to fabricate • These are small and are used to display characters	MP3 players, mobile phones
Active-matrix OLED (AMOLED)	AMOLEDs consist of full layers of cathode, organic material and anode. Thin-film transistors on anode are arranged in a matrix, which forms images with switching of pixels.	• Less power consumption • Faster refresh rate	Computers monitors, large TV screens and electronic sign boards
Transparent OLEDs (TOLEDs)	TOLEDs consist of transparent components—substrate, cathode and anode. When turned on, these allow light to pass through in bidirectional way. TOLEDs exhibit both PMOLED and AMOLED characteristics.	• Less power consumption • Good contrast, especially when exposed to bright sunlight	Headup displays, laptops, mobile phones and smart windows
Top-emitting OLEDs	Top-emitting OLEDs substrate is either opaque or reflective. This type of OLED is well suited with AMOLEDs applications due to integration of non-transparent transistor back pane.	• Higher luminance and colour purity than other OLEDs	Smart cards
Foldable OLEDs (FOLEDs)	Subtrate is made of flexible plastics or metallic foils. These OLEDs are used where the chances of breakage are high.	• Flexible and more durable • Less power consumption	GPS devices, mobile phones and large display devices (television sets)
White OLEDs	White OLEDs are constructed in large sheets. These emit white light and are brighter than fluorescent lights and incandescent bulbs.	• Less power consumption • Cost-effective and lightweight • Brighter than other OLEDs	Car lighting, flat lamps

TABLE II
MISCELLANEOUS PRODUCTS USING OLED TECHNOLOGY

Product name	Specifications
Royole Rowrite notepad 	It is a tablet like device manufactured by the Royole. It has an OLED screen with touch sensing.
Sony Aibo 2017 	It is an LTE Powered robot that aims to replicate a real dog. Aibo's eyes are small, round OLED displays that are used to blink and wink.
Sony HMS-3000 MT 	It is a 2D/3D head-mounted display for endoscopic imaging. It uses two Sony 1.8cm (0.7-inch) OLED micro displays and supports 3D.
Apogee Duet 2 	It is a second-generation audio interface for Mac with USB2.0 interface and full-colour OLED display.

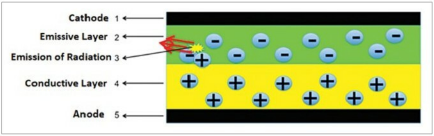

Fig. 2: Bilayer OLED

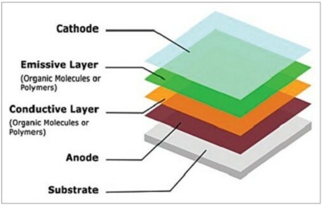

Fig. 3: OLED architecture

exhibit singlet and triplet exciton radiation phenomenon.

Architecture of an OLED

An OLED consists of:

1. Substrate. Regarded as the base of an OLED, it is made up of a thin translucent glass or foil material.

2. Anode. It is also called as emitter. Its main function is to emit electrons when a voltage is applied across terminals.

3. Organic layer. The layer above anode is called organic layer. It contains conductive polymer made of hydrogen or carbon molecules.

4. Conductive layer. This layer is made of organic plastic molecules and helps to move holes from anode.

5. Emissive layer. This layer is made of organic materials that are different from those used in the conductive layer. It helps to transport electrons from cathode.

6. Cathode. Cathode is the topmost part of OLED displays. It injects